Subproject 3:
Polishing a final product

Image and Video Analysis

G.E.E.T. 4th year

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- **General goal:** to analyze a video flow, detect an ID card shown to it, segment the card, analyze the different fields: the MRZ (machine readable zone), Face Photo, signature and personal data, save the name, ID number and Face picture in a dataset, and compare the Face picture with the selfie of the holder.

- **Application scenarios:**

  - **Static:** the ID-card is left on a surface below the camera and the ID-Card is analyzed when no movement is detected. A second static picture of the ID-Card held by the owner, showing also your face is analyzed for matching.

  - **Dynamic:** the ID-card is held in your hand in front of the camera and the ID-Card is analyzed as soon as a valid picture is taken. In this case the Selfie can be taken at the same type when showing the front or the bak of the card, or taken as a second (third) step after analyzing one or two sides.

- Wrapping up the product for a client
  - Product sheet: explanation of the product’s functionality
    - What does it do?
      - Does it work with any ID document? If not, what types of ID documents and what countries are supported?
      - What kind of data can it extract?
      - How long does it take to analyze a document from start to finish?
      - How many and what types of interactions are required from the user?
    - Operating conditions
      - Under what conditions does the product work as intended?
      - What hardware, sensors, operating system and libraries are needed to operate it?
      - What steps should the client follow to set up the product ?
      - What are the optimal scenario conditions (e.g., distance to the camera, lighting, background)
    - Troubleshooting: What Can Go Wrong?
      - List potential issues the client might encounter when setting up the product and provide possible solutions.
      - Identify problems the end user might face when trying to sign up for the service using this onboarding product and suggest likely causes and workarounds.
    - Pricing the Product (Not a Joke!)
      - Highlight what makes your product unique in the market and why the client should pay for it.
      - Consider the cost of developing the product compared to the problem the client would face if they chose another provider.

- Tasks
  - Dedicate most of your time to make a robust product: it is better to have less functions that always work that many fancy options that most of the time don't work.
  - Review and improve your code from parts 1 and 2.
  - If you have time and feel comfortable with your product go for the new functionalities →
  - Reading the data fields, at least the name and the ID number, robustly.
    - You can use external OCR libraries like EasyOCR or Tesseract-OCR
    - You can implement your own OCR by training from a dataset of printed letters (more time-consuming)
  - Matching the Face photo with the Selfie of the holder
    - You can use a external ad-hoc library like DeepFace
    - You can use OpenCV functions for face recognition or even train an ad-hoc one (more time consuming)
  - Liveness detection: making sure that the selfie is not a picture of the person in the ID-Card
    - For this "extra" functionality you can use also DeepFace.

What should you deliver for this part?

A code project with the next files:

- python file that runs in Windows platforms
- all the needed libraries to run the code
- The program should receive no arguments because it opens the webcam, but you should allow an optional argument to open a prerecorded videofile for testing
- The executable should analyze the frames from the videoflow and search for an ID document. As soon as a document is found, it should analyze the content and output the data found in it as explained before

A PowerPoint presentation explaining the method <u>and the Wrapping up document</u>

- you should specify how the program(s) can be used by a third party
- you should highlight the part of the code that has been copied and indicate where you copied it from, even if it comes from a forum discussion
- you should indicate under what conditions the method works and in what cases it doesn't work.
- you should include images to explain how the methods work and some example where it doesn't work and why.

What scores can you expect from Subproject 2?

Independently on the amount of work you do, your Project should fulfill the next requirements in order to pass this part:

1) The program should run in a system with Windows11, 64 bit, OpenCV4.X or upper, so make sure to give the correct instructions for me to run it from the source files as if I were a client. I advice you to simulate a second installation in a friend's computer. If you are using a library that is not provided by the distribution we use in class, provide it. 2) The executable should run as explained in the previous slide. Any modification of that should be explained and justified.

3) The PowerPoint should include the 4 points explained in the previous slide

4) The program should work (not halting!) for a video of any length or webcam flow.

5) The program should stop only after an ID is found in the video flow, the content is analyzed and, if the extra-functionalities are implemented, the selfie is taken.

The minimum functionality for every score range is the following (the individual score will depend on the contributions in the forum):

- Up to 5 points: If the program works as explained with any rotation of the ID in a flat surface and **extracts robustly** the image of the face and the signature in upright position
- Up to 7 points: if the program works as in 1) and also **segments robustly** the sub-images of the main personal data: DNI number, expiration date, name, surname, birthday and MRZ
- Up to 8 points if the program works as in 2) and also works when you **hold the ID with your hand** in front of the webcam
- Up to 9 points if the program works as in 2) but the name and ID-number is **robustly shown** in the console (OCR) and the selfie of he holder is taken and saved with the name, ID-number and Face photo in a **external dataset** (not processed).
- Up to 10 points if the program works as in 9 but the Face photo and the selfie are compared during the process, saving also in he dataset the result of the matching.
- Up to 10 points if the program works as in 9 but the selfie is analyzed to detect a fake shoot (a picture)